1. What is HTML5 and how is it different from previous versions of HTML?

Expected Answer: HTML5 is the latest version of HTML that introduces semantic elements, multimedia support, and APIs like local storage and canvas. Unlike older versions, it eliminates the need for external plugins (like Flash) and improves accessibility and structure.

2. What are semantic elements in HTML5? Give examples.

Expected Answer: Semantic elements clearly describe their meaning in the context of the page. Examples include <header>, <footer>, <article>, <section>, and <nav>. They improve readability, SEO, and accessibility.

3. What is the purpose of the <canvas> element in HTML5?

Expected Answer: <canvas> provides a drawable region in the browser where you can render graphics, animations, or games using JavaScript. It's commonly used for dynamic visual content.

4. How does HTML5 support multimedia content?

Expected Answer: HTML5 introduces <audio> and <video> tags that allow embedding media directly into web pages without third-party plugins. These tags support multiple formats and controls.

5. What is the difference between <section> and ?

Expected Answer: <section> is a semantic element used to group related content with a heading, while is a generic container with no semantic meaning. Use <section> when the content has a thematic grouping.

6. What is the role of the <!DOCTYPE html> declaration?

Expected Answer: It tells the browser to render the page in standards-compliant mode using HTML5. It ensures consistent behavior across browsers.

7. How does HTML5 handle form validation?

Expected Answer: HTML5 introduces built-in form validation attributes like required, pattern, min, max, and type="email" or type="number". These reduce the need for JavaScript-based validation.

8. What are data attributes in HTML5 and how are they used?

Expected Answer: Data attributes (data-*) store custom data in HTML elements. They're useful for passing metadata to JavaScript without affecting the DOM structure. Example: .

9. What is the difference between <article> and <aside>?

Expected Answer: <article> represents independent, self-contained content (like a blog post), while <aside> contains tangential content (like a sidebar or related links).

10. How does HTML5 improve accessibility?

Expected Answer: Semantic tags help screen readers and assistive technologies understand page structure. Combined with ARIA roles and proper labeling, HTML5 enhances user experience for all users.

11. What is the purpose of the <meta> tag in HTML5?

Expected Answer: <meta> provides metadata about the document, such as character encoding (<meta charset="UTF-8">), viewport settings, and SEO-related descriptions.

12. Can you explain the use of the <picture> element in HTML5?

Expected Answer: <picture> allows responsive image loading by specifying multiple sources for different screen sizes or resolutions. It helps optimize performance and display quality.

13. What is the difference between and tags?

Expected Answer: emphasizes text (usually italic), while indicates strong importance (usually bold). Both have semantic meaning and aid accessibility.

14. How does HTML5 support offline storage?

Expected Answer: HTML5 provides localStorage and sessionStorage APIs to store data in the browser. This enables offline access and persistent state across sessions.

15. What are some common accessibility best practices when writing HTML5?

Expected Answer:

- Use semantic tags for structure
- Add alt attributes to images
- Label form fields with <label>
- Ensure keyboard navigation is supported